Khang Tran

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Education

Princeton University, BA in Physics; minors in Computer Science and Engineering – Princeton, NJ — Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Computer Graphics, Neural Rendering, Machine Learning & Pattern Recognition, Algorithms & Data Structures, Honors Linear Algebra, Multivariable Calculus

NUS High School of Mathematics and Science, Diploma (High Distinction) with Honours in Engineering & Computer Science, and Majors in Mathematics, Chemistry & Physics – Singapore — Jan 2020 – Oct 2023

- **GPA:** 4.9/5.0 — Computer Organization, Operating Systems (NUS dual enrollment)

Experience

Undergraduate Researcher, Roberto Car Group, Princeton University – Princeton, NJ — June 2026 – present

- Built an auto-relaunching pipeline to run large-scale physics (DFT) simulations across multi-node HPC (SLURM) clusters, automatically recovering from job timeouts and failures.
- Automated job orchestration and parallelization across nodes to scale throughput.

Space Systems Engineer Intern, Aliena – Singapore — Oct 2023 – Mar 2024

- Built an end-to-end Python software pipeline for real-time control, data acquisition, and visualization of laboratory instrumentation.
- Designed and built a hardware-in-the-loop tester – a programmable board emulating a Hall thruster's signals across nominal and fault modes – to exercise the control-unit (ECU) software without live hardware in the loop.

Projects

2D-to-3D Style Transfer for Reusable Assets — Feb 2026 – May 2026

COS526: Neural Rendering (Princeton, Prof. Felix Heide)

- Developed a pipeline that turns a 3D mesh and a single style image into a fully stylized, game-ready asset with physically based rendering (PBR) textures (albedo, normal, roughness) baked via differentiable rendering.
- Built the multiview-consistent style-transfer stage — stylized 16 Kaolin-rendered views with SDXL + ConsisLoRA and depth/Canny ControlNets, enforcing cross-view consistency via cumulative view-warping, histogram matching, BiRefNet masking, and a RePaint inpainting schedule.

Shielding-Free Signal-Noise Suppression in Portable Low-Field MRI – Singapore — Apr 2022 – Aug 2023

Singapore University of Technology and Design (SUTD)

- Developed a signal- and image-processing system to suppress noise in SUTD's portable MRI (advised by Prof. Huang Shaoying); **Silver**, Singapore Science & Engineering Fair 2023, and poster at ICMRM 2023.

Skills

Programming: C/C++, Python, GLSL/WebGL, Git, MATLAB, Mathematica, LaTeX; data structures & algorithms (COS226)

Graphics & ML: Real-time rendering, ray tracing, shaders, image/signal processing (COS426); neural rendering & novel-view synthesis (COS526); PyTorch, GPU computing

Awards: Pyka Prize for Physics Excellence, Princeton Physics Dept. (2026); IBM Qiskit Hackathon **Champion COSCON 2025 1st** (Undergraduate); Singapore Physics Olympiad Gold (2022)

Leadership & Teaching: CS Lab TA, COS217/226 (2026) & Physics Tutor, Princeton Physics Dept. (2025–26); Problem Setter & Organizer, Princeton Science Olympiad (2025–26) & Singapore Physics League / SPhL (2023–25)